CHAPTER 7

Antiderivatives and differential equations

S52, S53, S55, S58

Before you begin · Differentiation and models

Without a calculator, in six minutes. Answer briefly. Use the derivative, the exponential and the units. Then choose the targeted follow-up indicated under the questions.

1. Differentiate $F(x) = x^2$.

2. Find an affine function whose derivative is 3.

3. Differentiate 3 e^{2x}.

4. In $f(t) = 80e^{-0,1t}$, calculate $f(0)$.

5. Convert 90 min to hours.

6. What is e^0 ?

Answers

$F' = 2x$; $3x + C$; $6e^{2x}$; 80 ; 1,5 h ; 1.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Go back to the derivative of x squared. **2.** Write mx+b and differentiate to determine m. **3.** Identify the factor 2 from the composition; R36. **4.** Substitute t=0 before calculating the exponential. **5.** Divide the minutes by 60; R67. **6.** Recall the value of exp at 0. The R questions serve as material for review with the teacher, even if their whole-class routine is planned later.

Objectives

Calculate antiderivatives; solve a differential equation with initial condition; interpret an Euler approximation.

Course notes

Antiderivatives on an interval

We assume at this stage that every function continuous on an interval has antiderivatives; the link with area will be established in chapter 8. An antiderivative of f over an interval I is a differentiable function F such that $F' = f$ on I. If F is an antiderivative, all the antiderivatives on I are $F + C$, $C \in \mathbb{R}$.

Proof. If $F' = G' = f$, then $(F - G)' = 0$ on the interval; $F - G$ is therefore constant. Conversely, $(F + C)' = f$. The interval assumption is essential: on a disconnected domain, the constants may differ from one interval to another.

Table of antiderivatives - read the derivatives table backwards

$f(x)$	$F(x)$
$x^n \ (n \in \mathbb{Z},\ n \neq -1)$	$\frac{x^{n+1}}{n+1}$
$\frac{1}{x}$	$\ln \lvert x \rvert$ on an interval avoiding 0
$\frac{1}{\sqrt{x}}$	$2\sqrt{x} \quad (x > 0)$
e^x	e^x
$\cos x$	$\sin x$
$\sin x$	$-\cos x$

The additive constant can always be added. The last two lines will be reused after studying trigonometry.

Recognising a composition

For a function u differentiable, the antiderivatives of $u'u^n$, $u'e^u$, $\frac{u'}{u}$ are respectively $\frac{u^{n+1}}{n+1}$ $(n \neq -1)$, e^u, $\ln \lvert u \rvert$, on intervals where the expressions are defined. Linearity handles sums.

Example. An antiderivative of $6\,x(x^2+1)^2$ is $(x^2+1)^3$; an antiderivative of e^{3x} is $\frac{e^{3x}}{3}$. To impose $F(0) = 2$ in the latter case, take $F(x) = \frac{e^{3x}}{3} + \frac{5}{3}$.

Differential equations

A differential equation links an unknown function to its derivatives. A solution must be checked on a specified interval.

Standard solutions

$y' = ay$ has solutions $y(x) = Ce^{ax}$. For $a \neq 0$, $y' = ay + b$ has solutions

$$y(x) = Ce^{ax} - \frac{b}{a}.$$

If $a = 0$, the equation becomes $y' = b$ and $y(x) = bx + C$. If g is a known particular solution of $y' = ay + f(x)$, the solutions are $y = g + Ce^{ax}$.

Proof for $y' = ay$. If y is a solution, $z(x) = e^{-ax}y(x)$ satisfies $z' = e^{-ax}(y' - ay) = 0$; $z = C$. Conversely, the derivative of Ce^{ax} is aCe^{ax}. For $y' = ay + b$, subtract the constant solution $-\frac{b}{a}$ reduces to the homogeneous equation. For a variable right-hand side, do the same with g.

Example. $y' = -2y + 6$ and $y(0) = 1$ give $y = 3 + Ce^{-2x}$, then $C = -2$. The solution $3 - 2e^{-2x}$ increases to the equilibrium 3 when $x \to +\infty$.

Euler's method

For $y' = ay + b$, with a step $h > 0$, we approximate $y(t_k)$ by Y_k via $t_{k+1} = t_k + h$, $Y_{k+1} = Y_k + h(aY_k + b)$. This is an approximation using the tangent, not an equality for the exact solution. A smaller step can improve the approximation; check the error numerically on a known.

Differentiating backwards and modelling change

7.1 Recognising and checking an antiderivative

First use: S52.

Definition · Antiderivative on an interval

A function F is an antiderivative of f on I if F is differentiable on I and $F' = f$ on I. The interval is part of the data. Finding an antiderivative is an inverse calculation; differentiation makes it possible to check it.

Worked example · Correcting a factor

For $f(x) = 3e^{2x}$, $F(x) = \frac{3}{2}e^{2x}$ works because $F'(x) = \frac{3}{2} \times 2e^{2x} = 3e^{2x}$. For $g(x) = 2x/(1+x^2)$ on $\mathbb{R}$, the antiderivative $G(x) = \ln(1+x^2)$ uses a positive argument. All antiderivatives are obtained by adding a constant.

A1 · Immediate application Determine an antiderivative of 4 $x^3 - 2x + 1$, of e^{-3x} and of 1 $/(2x+1)$ on a suitable interval. Check by differentiation.

7.2 A constant, a vertical translation

First use: S52.

Proof. If $F' = G' = f$ on an interval I, then $(F-G)' = 0$ on I. A function whose derivative is zero on an interval is constant; therefore $F - G = C$. On a domain made of several disjoint intervals, the constant may differ on each interval. □

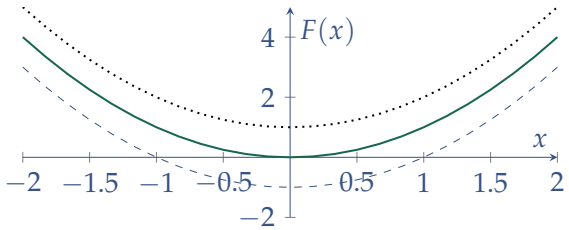


Figure 7.1. Three antiderivatives of $2x$: at the same x-values, the tangents have the same slopes.

Worked example · One value fixes the translation

The antiderivatives of $2x$ are $x^2 + C$. Imposing F(0)=-1 selects $F(x) = x^2 - 1$. The graph is translated one unit downwards without changing the slopes.

A2 · Partial guidance Determine the antiderivative of $2x$ which equals 3 at 1. Then explain why all the antiderivatives of $1/x$ on $\mathbb{R} \setminus \{0\}$ are not described by a single constant common to both half-lines.

7.3 The unknown is a function

First use: S55.

Property · Equation $y' = ay$

For $a \in \mathbb{R}$, the solutions on $\mathbb{R}$ are $y(t) = Ce^{at}$, with $C \in \mathbb{R}$. An initial value $y(0) = y_0$ imposes $C = y_0$.

Proof. A function of this form is indeed a solution. Conversely, if y is a solution, $z(t) = y(t)e^{-at}$ has derivative $(y' - ay)e^{-at} = 0$, so it is constant on the interval. Thus $y(t) = Ce^{at}$. The proof checks the solutions and shows that none is missing. □

Worked example · A condition at another time

For $y' = 3y$ and y(1)=2, $y(t) = Ce^{3t}$, then $Ce^3 = 2$. Hence $C = 2e^{-3}$ & $y(t) = 2e^{3(t-1)}$. The constant equals the prescribed value only if that value is given at 0.

A3 · Immediate application Solve $y' = -2y$ with $y(0) = 5$. Is the number 0 a possible solution of the differential equation alone? Of this initial value problem?

7.4 Subtract a known solution

First use: S55 and S58.

Worked example · Variable right-hand side

For $y' = -y + 2t + 1$, look for a particular solution $p(t) = at + b$ leads to $a = -at - b + 2t + 1$. Identifying the coefficients gives $a = 2$, $b = -1$. If y is a solution, $z = y - p$ satisfies $z' = -z$. With $y(0) = 3$, we get $z(0) = 4$, so $y(t) = 2t - 1 + 4e^{-t}$.

A4 · Partial guidance Solve $y' = 2y - 4$ with $y(0) = 3$ by subtracting a constant solution. Check the formula and initial value.

A5 · Fix a proof A student checked that a function satisfies a differential equation, then wrote "so I have found all the solutions". Why is this conclusion insufficient? Illustrate with $y' = y$.

7.5 Comparing the discrete, the continuous and an approximation

First use: S55 and S58.

Worked example · Cooling

The model $T' = -0{,}1(T - 20)$, with $T(0) = 80$ and t in minutes, gives $T(t) = 20 + 60e^{-0{,}1t}$. With a step h, Euler's method gives $T_{n+1} = T_n - 0{,}1h(T_n - 20)$ at the times $t_n = nh$. For $h = 2$, we get 80, 68, 58.4. These are approximations of the continuous model at times 0, 2, 4; a discrete physical law given independently is not automatically an Euler scheme.

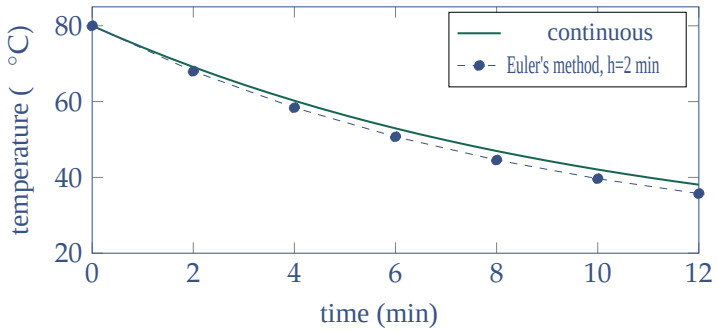


Figure 7.2. Read the gap at t=6 min; the segments join the successive approximations.

A6 · Choice of strategy Determine the first time the continuous model reaches 50 degrees Celsius. Compare with the first value from Euler's method, with h=2 min, which is less than or equal to 50. Explain why the two results differ.

A first step: First solve an exponential equation; for Euler, write the formula for $T_n - 20$.

One minute, spoken aloud. State what y, y' and t denote, then check that the units of both sides of a differential equation are compatible.

Delayed review · S78. Complete the table of antiderivatives with trigonometric functions and check the factors of the compositions.

Investigation

Cooling a drink - 40 min.

A drink initially at 80 °C is placed in a room kept at 20 °C. For $t \geq 0$, let $T(t)$ be its temperature in degrees Celsius, t denoting the time elapsed in minutes. We assume the model $T'(t) = -0{,}1(T(t) - 20)$ and $T(0) = 80$.

1. Solve this initial value problem and give the expression of $T(t)$.

2. Study the direction of variation of T and determine its limit at $+\infty$. Interpret the results.

3. Determine the first time the temperature is 40 °C. Give the exact value, then an approximate value in minutes.

4. Apply two steps of Euler's method with a step of 1 minute. Compare the approximation obtained after two minutes with the exact value $T(2)$.

5. Give a concrete situation in which the assumption of a constant ambient temperature would no longer be suitable.

Progressive exercises

E1 · Direct application. Determine all antiderivatives on $\mathbb{R}$ of the function $x \mapsto 3x^2 - 4x + 1$.

E2 · Application and justification. Determine the antiderivative F of the function $x \mapsto \frac{2x}{x^2+4}$ on $\mathbb{R}$ which satisfies $F(0) = 0$.

E3 · Method to organise. Determine the solution on $\mathbb{R}$ of the differential equation $y' = 3y$ that satisfies $y(1) = 2$.

E4 · Choice of strategy. Determine the solution on $\mathbb{R}$ of the differential equation $y' = -y + 4$ which satisfies $y(0) = 7$.

E5 · Synthesis or proof. Consider the differential equation $(E) : y' = y - x$ on $\mathbb{R}$. Check that $g(x) = x + 1$ is a solution of it, then find the solution of (E) which satisfies $y(0) = 0$.

Work from the textbook or a past paper

Barbazo, p. 228, Exercises 25 and 30. In S55: exercise 30, to bring out a constant solution.
At home: exercise 25 in full, distinguishing a condition on f from a condition on f'.

Homework

DM7 - Change with a variable contribution - 60 min.

We look for a function y defined and differentiable on $[0; +\infty[$ such that $y'(t) = -y(t) + 2t + 1$ and $y(0) = 3$.

1. Find two real numbers a and b so that $g(t) = at + b$ is a solution of the differential equation.

2. Deduce the expression of the solution satisfying $y(0) = 3$.

3. Study the variations of this solution and state its minimum and the time at which it is reached.

4. Write a Python function with argument N, a strictly positive integer, that applies Euler's method on $[0; 2]$ with the step $\frac{2}{N}$ and returns the approximation of $y(2)$. Compare with the exact value for $N = 10$, then $N = 100$.

5. Why is checking the differential equation at only a few instants not enough to prove that a function is a solution?

MCQ A · families of antiderivatives

After S53; MCQ B after S58.

For each question, only one option is correct. Justify your choice orally before checking the solutions.

1. An antiderivative of x^2 on $\mathbb{R}$ is:

A. $2x$ B. x^3

C. $x^3/3$ D. $3x^3$

2. All antiderivatives of e^{2x} are:

A. $e^{2x} + C$ B. $\frac{1}{2}e^{2x} + C$

C. $2e^{2x} + C$ D. $xe^{2x} + C$

3. An antiderivative of $2x/(1 + x^2)$ on $\mathbb{R}$ is:

A. $\frac{1}{1+x^2}$ B. $2\ln x$

C. $\ln x$ D. $\ln(1 + x^2)$

4. Two antiderivatives of the same function on an interval have graphs:

A. images of each other under a vertical translation B. necessarily identical

C. always intersecting D. images under any horizontal translation

5. The antiderivative of $2x$ satisfying $F(1) = 3$ is:

A. $x^2 + 3$ B. $2x^2 + 1$

C. $x^2 + 2$ D. $x^2 - 2$

MCQ B · differential equation and model

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. The solution of $y' = -2y$, $y(0) = 3$, is:

A. $3e^{2t}$ B. $3e^{-2t}$

C. $-2e^{3t}$ D. $3 - 2t$

7. The constant solution of $y' = -2y + 6$ is:

A. 6 B. -3

C. 2 D. 3

8. If y_p is a solution of $y' = ay + b(t)$, then $z = y - y_p$ satisfies:

A. $z' = az$ B. $z' = az + b(t)$

C. $z' = b(t)$ D. $z' = 0$ necessarily

9. Euler with step h for $y' = -y$, from $y_0 = 2$, gives:

A. $y_1 = 2 + h$ B. $y_1 = e^{-h}$

C. $y_1 = 2 - 2h$ D. $y_1 = 2 - h^2$

10. If t is in minutes and T in degrees Celsius, the coefficient k of $T' = -k(T - 20)$ is expressed in:

A. minutes B. reciprocal minutes

C. degrees Celsius D. litres

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. An antiderivative is unique without any further condition.

2. Euler's method gives the exact value for every differential equation.

3. Checking a solution consists of checking the equation and the initial condition.

MCQ solutions

1. **C.** d/dx of $x^3/3$ gives back x^2. Differentiate x^2 is not finding an antiderivative.

2. **B.** The factor 1/2 compensates for the factor 2 from differentiating the composite. C is a real constant.

3. **D.** This is the form u'/u, with $u = 1 + x^2 > 0$ everywhere. $\ln x$ is not defined on all $\mathbb{R}$.

4. **A.** The difference between antiderivatives is constant; this translates into a vertical translation.

5. **C.** The antiderivatives are $x^2 + C$. The condition gives $1 + C = 3$, so $C = 2$.

6. **B.** The solutions are Ce^{-2t} ; the initial condition gives $C = 3$. An affine function is not suitable at every instant.

7. **D.** A constant has zero derivative; solve $0 = -2L + 6$ gives $L = 3$.
8. **A.** Subtracting the two equations eliminates the right-hand side. This is the associated homogeneous equation.
9. **C.** The update is $y_1 = y_0 + h(-y_0) = 2 - 2h$. Euler gives an approximation, distinct from the exact solution $2e^{-h}$.
10. **B.** The derivative is in degrees Celsius per minute: k multiplies a temperature difference and has unit min^{-1}.

True/false. False: adding a constant gives another antiderivative. False: for $y' = -y$, $y(0) = 2$, a step of 1 gives 0, instead of $2/e$. True, on the required interval and with the required differentiability.

Solutions to the applications and guidance

Application 1. $x^4 - x^2 + x$; $-e^{-3x}/3$; $\frac{1}{2}\ln|2x+1|$ on each of the intervals not containing $-1/2$. Differentiating gives back the stated expressions.

Application 2. $F = x^2 + C$ and $F(1) = 3$ give C=2. On the two half-lines, $\ln|x| + C_-$ for $x < 0$ and $\ln|x| + C_+$ for $x > 0$ is an antiderivative, even if $C_- \neq C_+$.

Application 3. $y = 5e^{-2t}$. The zero function satisfies the equation alone, but not the condition $y(0) = 5$.

Application 4. The constant 2 is a solution. $z = y - 2$ satisfies $z' = 2z$ and $z(0) = 1$, therefore $y = 2 + e^{2t}$. Its derivative $2e^{2t}$ is indeed 2 $y - 4$ and $y(0) = 3$.

Application 5. Check e^t only shows that it is a solution. The functions 2 e^t, $-e^t$ and the zero function are others. For completeness, start from an arbitrary solution and show that it has the stated form.

Application 6. Continuous: $e^{-0{,}1t} = 1/2$, i.e. $t = 10\ln 2 \simeq 6{,}93$ min. Euler's method: $T_n = 20 + 60(0{,}8)^n$; $(0{,}8)^3 = 0{,}512 > 1/2$, $(0{,}8)^4 = 0{,}4096 < 1/2$. The first grid point is therefore at 8 min. The step imposes discrete times and Euler's method introduces an approximation error.

Research and comparison

Have students calculate a first step by hand, then compare h=2 and h=1 with the given program.
The observed convergence of an approximation does not replace an estimate of its error.
Do not call a list of Euler values an "exact solution".

Teaching notes and solutions

Guidance for the teacher

Distinguish the unknown function from the real variable. Have each antiderivative checked by differentiation. For $y' = ay + b$, first seek the constant solution. The case $a = 0$ must remain visible. The trigonometric antiderivatives are given here as a reference table, but their assessed use must wait for the trigonometry unit. For Euler's method, keep the units and compare with a known exact solution.

Solutions to the activity

1. The equation is $T' = -0{,}1T + 2$. A constant solution is 20; all solutions are $T(t) = 20 + Ce^{-0{,}1t}$. Since $T(0) = 80$, $C = 60$. We check $T'(t) = -6e^{-0{,}1t} = -0{,}1(T(t) - 20)$ and $T(0) = 80$.

2. $T'(t) = -6e^{-0{,}1t} < 0$: the temperature strictly decreases. Since $e^{-0{,}1t} \to 0$ at $+\infty$, $T(t) \to 20\,°\text{C}$, remaining above 20 °C.

3. $T(t) = 40 \iff e^{-0{,}1t} = \frac{1}{3}$. Taking the logarithm, $-0{,}1t = -\ln 3$, i.e. $t = 10\ln 3 \simeq 10{,}9861$ min. Strict decrease ensures that this is the only time it passes through 40 °C.

4. For $h = 1$, $Y_{k+1} = Y_k - 0{,}1(Y_k - 20)$. Thus $Y_1 = 80 - 6 = 74$ and $Y_2 = 74 - 5{,}4 = 68{,}6$. The exact solution gives $T(2) = 20 + 60e^{-0{,}2} \simeq 69{,}1238$. Here Euler underestimates the temperature by about $0{,}5238\,°\text{C}$.

5. The model assumes an ambient temperature of 20 °C and a constant exchange coefficient of $0{,}1\ \text{min}^{-1}$. A change of room, a lid or stirring may alter these assumptions; the formula is then no longer automatically appropriate.

E1. Linearity gives $F(x) = x^3 - 2x^2 + x + C$, $C \in \mathbb{R}$. We check $F'(x) = 3x^2 - 4x + 1$. All antiderivatives on the interval $\mathbb{R}$ differ by a constant.

E2. The denominator is positive and its derivative is $2x$. The antiderivatives are $\ln(x^2+4) + C$. The condition at 0 gives $C = -\ln 4$, so $F(x) = \ln(x^2+4) - \ln 4$; we check $F(0) = 0$ and $F'(x) = \frac{2x}{x^2+4}$.

E3. The solutions of $y' = 3y$ are Ce^{3x}. The condition $Ce^3 = 2$ requires $C = 2e^{-3}$. Thus $y(x) = 2e^{3(x-1)}$, whose derivative is $3y(x)$ and whose value at 1 is 2.

E4. The constant 4 is a particular solution. The solutions are therefore $4 + Ce^{-x}$. Since $4 + C = 7$, $C = 3$. The solution is $y(x) = 4 + 3e^{-x}$; its derivative $-3e^{-x}$ is indeed $-y + 4$, and $y(0) = 7$.

E5. $g(x) = x + 1$ satisfies $g' = 1 = g - x$. If $z = y - g$, then $z' = z$, therefore $y = x + 1 + Ce^x$. The condition $y(0) = 0$ requires $C = -1$. We obtain $y = x + 1 - e^x$ and we check $y' = 1 - e^x = y - x$.

Solutions to homework DM7

1. If $g(t) = at + b$, then $g' = a$ and $-g + 2t + 1 = (2 - a)t + 1 - b$. For the identity to hold for every t, we need $2 - a = 0$ and $a = 1 - b$. Hence $a = 2$, $b = -1$, and $g(t) = 2t - 1$.

2. Let $z = y - g$. By subtracting the two equations, $z' = -z$, therefore $z = Ce^{-t}$. Thus $y(t) = 2t - 1 + Ce^{-t}$. Since $y(0) = 3$, $C = 4$. The function $y(t) = 2t - 1 + 4e^{-t}$ does satisfy $y' = 2 - 4e^{-t} = -y + 2t + 1$ and $y(0) = 3$.

3. $y'(t) = 2 - 4e^{-t}$. This derivative is negative for $t < \ln 2$, zero at $\ln 2$, positive after. The solution decreases on $[0, \ln 2]$ then increases on $[\ln 2, +\infty[$. Its minimum is $y(\ln 2) = 2\ln 2 - 1 + 2 = 1 + 2\ln 2$.

4.
```
def euler_dm7(N):
    h = 2/N
    y = 3.0
    for k in range (N):
        t = k*h
        y = y+h*(-y+2*t+1)
    return y
```

The time used on the right-hand side is the time at the start of the step. The program gives $Y_{10} \simeq 3{,}42950$ and $Y_{100} \simeq 3{,}53048$, whereas $y(2) = 3 + 4e^{-2} \simeq 3{,}54134$. The absolute errors are about 0 ,11184 and 0,01086. To check these values without a loop, let $Z_k = Y_k - (2kh - 1)$. The Euler's recurrence relation gives $Z_{k+1} = (1-h)Z_k$ and $Z_0 = 4$. Hence $Y_N = 3 + 4(1 - \frac{2}{N})^N$.

5. A solution of a differential equation must satisfy an equality between functions on the whole interval, as well as the initial condition. A few numerical values only check a finite number of points; the symbolic substitution in question 2 is a check over the whole domain.

Solutions to the textbook or past-paper work

Exercise 30. If y is constant equal to ℓ, the equation becomes $0 = -\frac{1}{2}\ell + 3$, so $\ell = 6$. Setting $z = y - 6$, we get $z' = -\frac{1}{2}z$; all solutions are $y(x) = 6 + Ce^{-\frac{x}{2}}$, $C \in \mathbb{R}$.

Exercise 25.

1. $f' = \frac{1}{2}f$ gives $f(x) = Ce^{\frac{x}{2}}$. Since $f(0) = 1$, $C = 1$.
2. $5f' = f$ gives $f(x) = Ce^{\frac{x}{5}}$. The condition $f(1) = 0$ requires $Ce^{\frac{1}{5}} = 0$; since the exponential is never zero, $C = 0$. The solution is the zero function.
3. $f' + f = 0$ gives $f(x) = Ce^{-x}$, therefore $f'(x) = -Ce^{-x}$. The condition relates to the derivative: $f'(1) = 2$ is equivalent to $-Ce^{-1} = 2$, i.e. $C = -2e$. Thus $f(x) = -2e^{1-x}$.